

Introduction to Computer Architecture

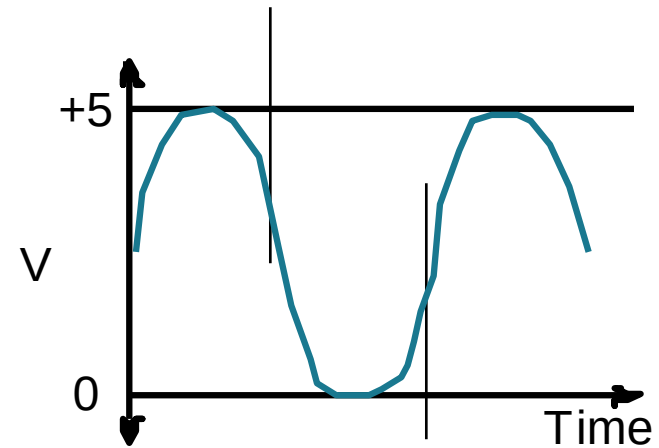
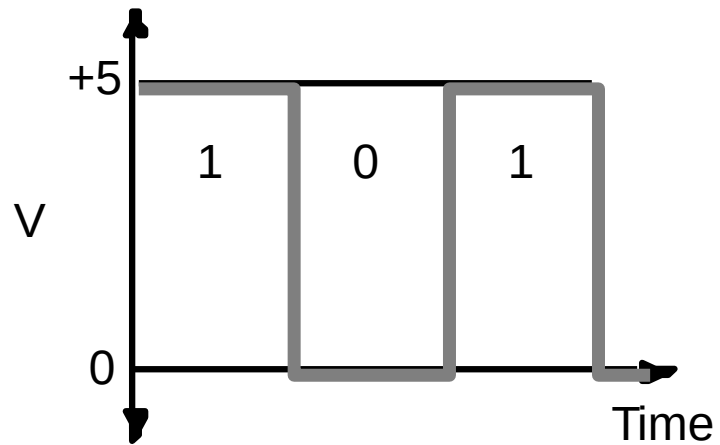
Digital Logic Circuits

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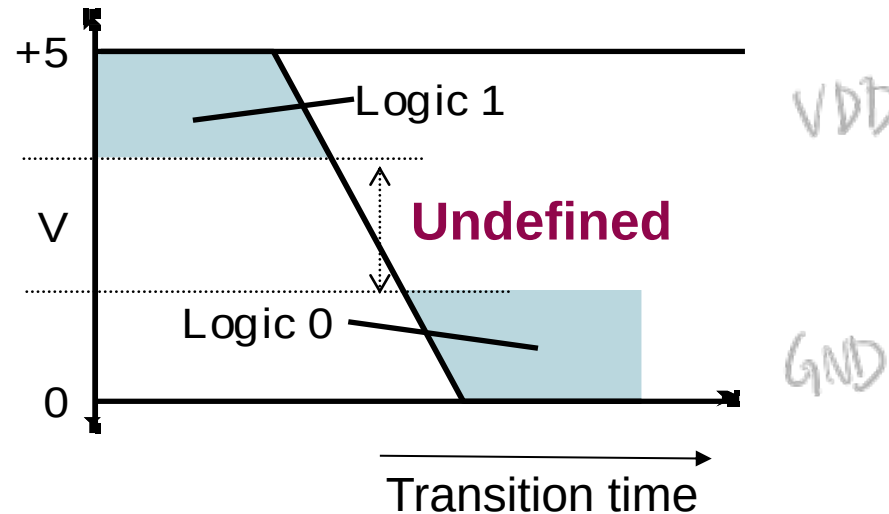
Digital Signals

■ Digital vs. Analog Waveforms



Digital Circuit

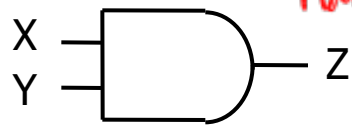
- A signal above a certain voltage level: “1” (True)
- A signal below a certain voltage level: “0” (False)



- Different components in a computer have different voltage levels
 - ❖ CPU (Core 2 Duo): 1.325 V
 - ❖ Chipsets: 1.45 V
 - ❖ Peripheral devices: 3.3V, 1.5V

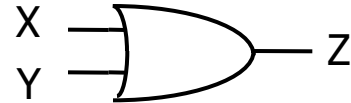
Boolean Algebra and Logical Operations

X	Y	X AND Y
0	0	0
0	1	0
1	0	0
1	1	1

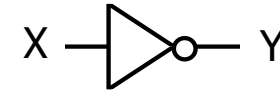


Hardware component
gate ←

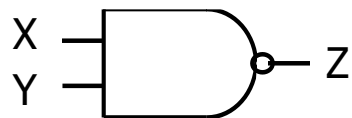
X	Y	X OR Y
0	0	0
0	1	1
1	0	1
1	1	1



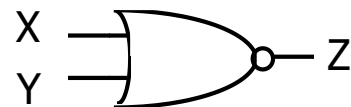
X	NOT X
0	1
1	0



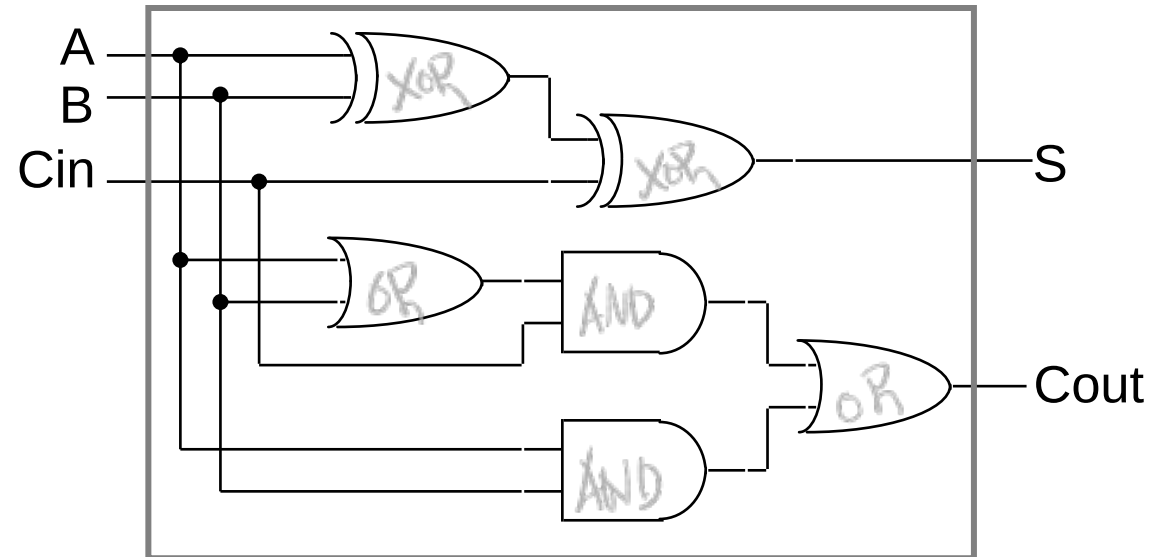
X	Y	X NAND Y
0	0	1
0	1	1
1	0	1
1	1	0



X	Y	X NOR Y
0	0	1
0	1	0
1	0	0
1	1	0

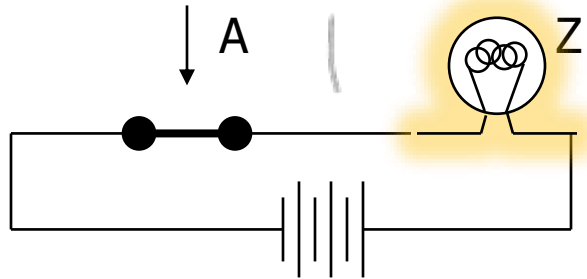


Digital Logic Circuit Example

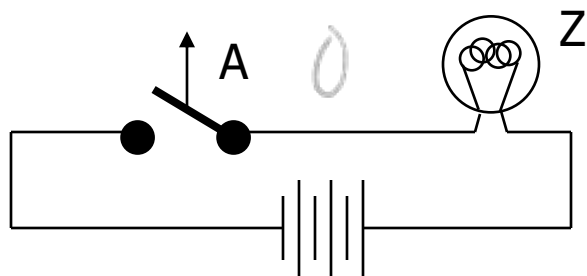


Switches: Basic Element of Physical Implementations

- Implementing a simple circuit



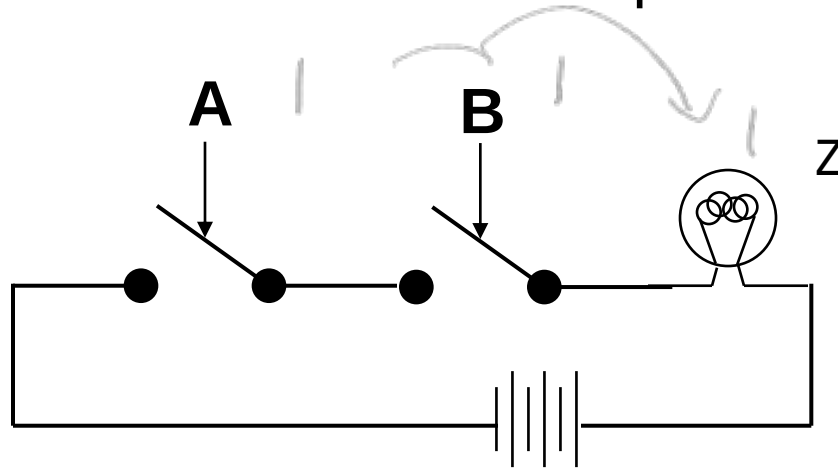
close switch (if A is "1")
and turn on light bulb (Z)



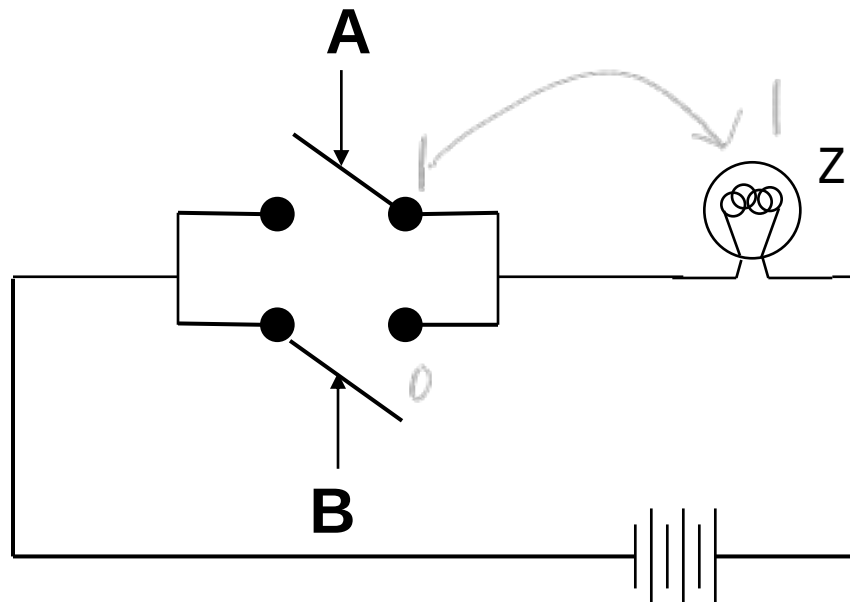
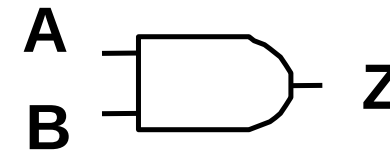
open switch (if A is "0")
and turn off light bulb (Z)

Switches (cont'd)

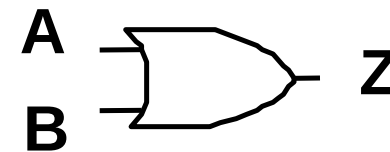
- Compose switches into more complex ones (Boolean functions):



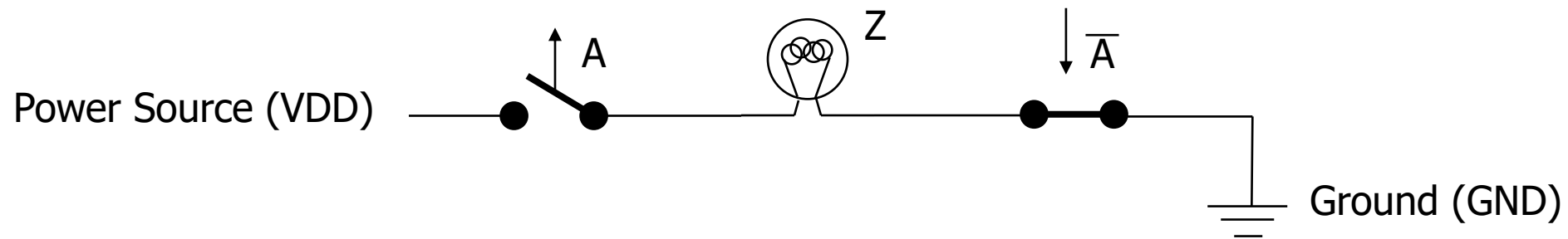
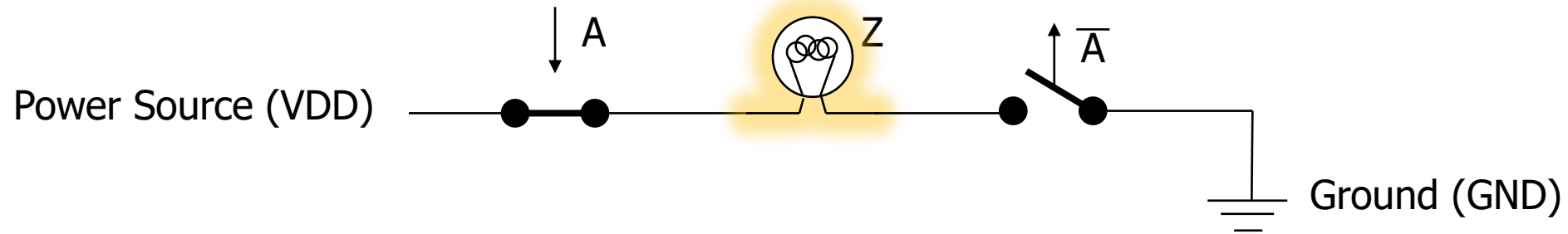
$$Z = A \text{ AND } B$$



$$Z = A \text{ OR } B$$

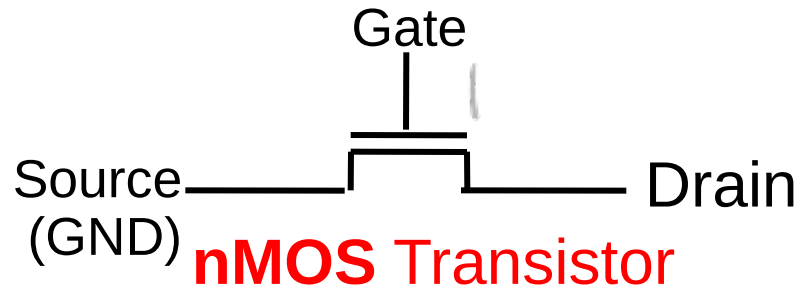


Semiconductor Switches

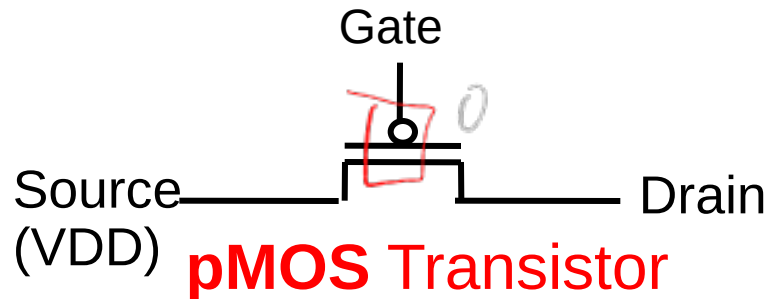


MOS transistors

- MOS transistors have three terminals: drain, gate, and source



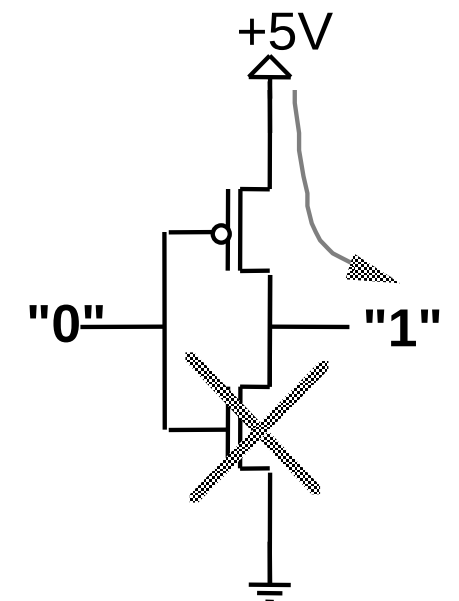
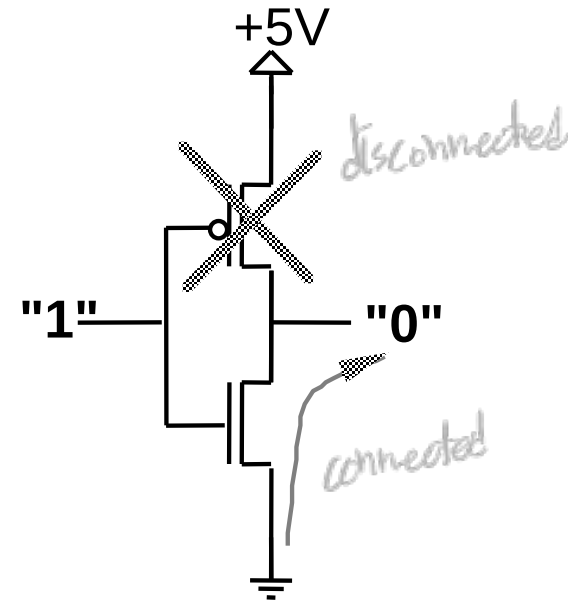
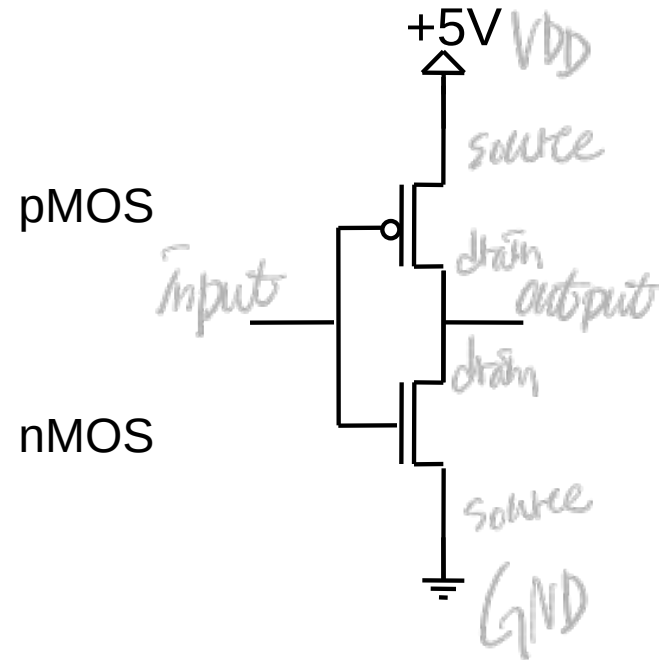
Logic 1 on gate:
Source and Drain connected



Logic 0 on gate
Source and Drain connected

- MOS transistors act as voltage-controlled switches

Inverter (NOT Gate) Operation



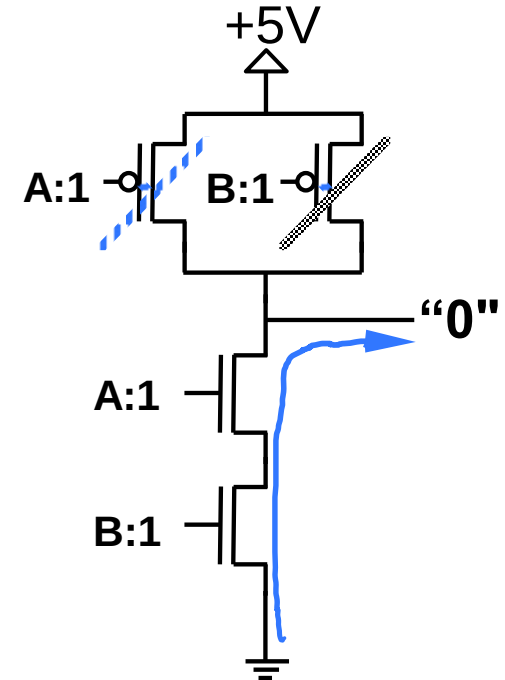
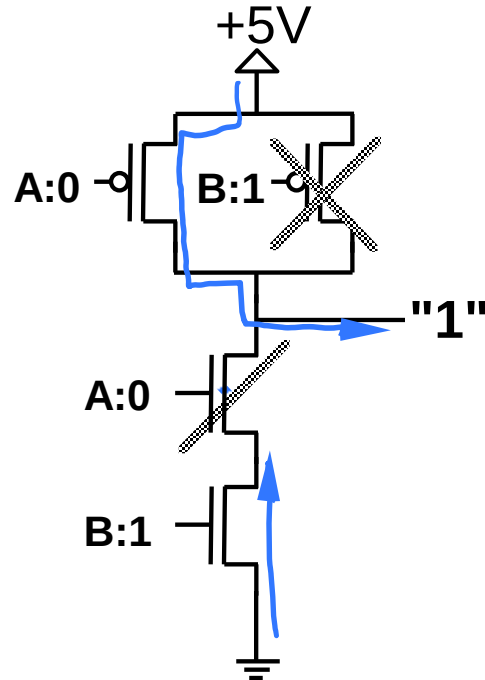
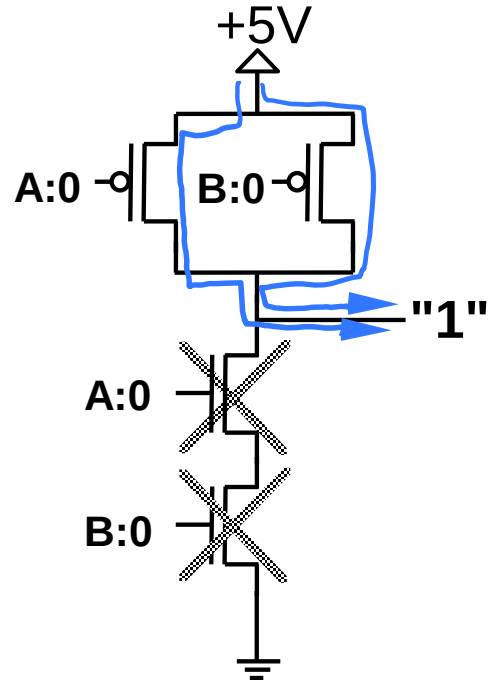
negate the input
as the output

NAND Operation

^A X	^B Y	X NAND Y
0	0	1
0	1	1
1	0	1
1	1	0

pMOS

nMOS



Number Systems - Binary Number

■ Binary numbers

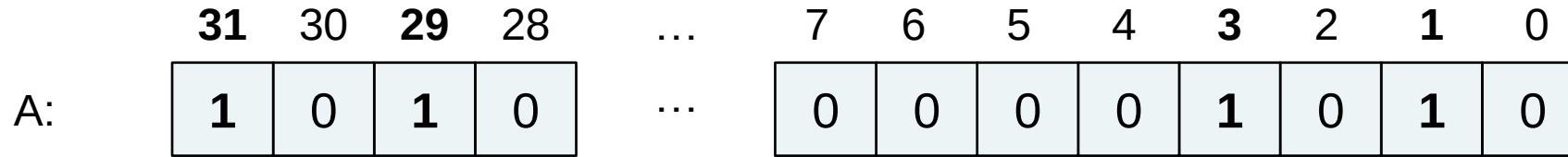
- ❖ Bit represents one of 2 values: 0 or 1
- ❖ Each digit of a binary number has 2x the weight of the previous digit

➤ ex) $10110_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 22_{10}$

- ❖ N-bit binary number represents one of 2^N possibilities
ex) 3-bit binary number represents one of 8 possibilities: 0 ~ 7

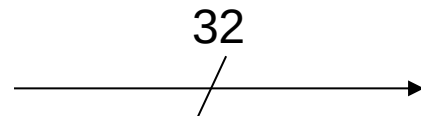
32-Bit Binary Number

- 32bit positive integer

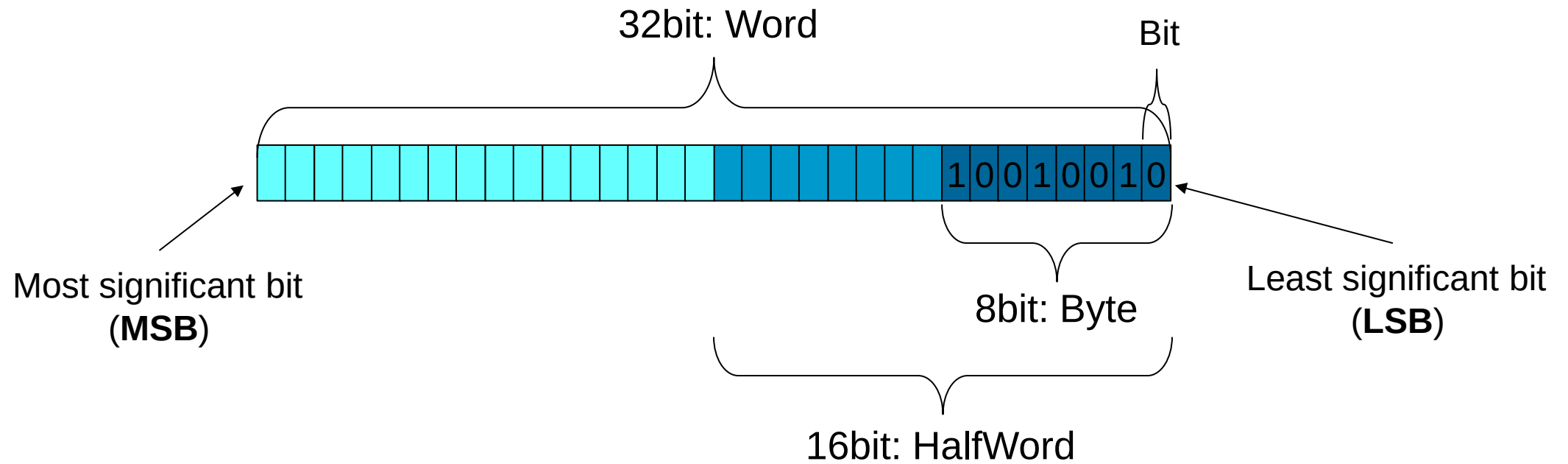


$$2^{31} + 2^{29} + 2^3 + 2^1$$
$$= 2,147,483,648 + 536,870,912 + 8 + 2 = 2,684,354,570$$

- Notation for a 32-bit wide data: A[31:0]



Bits, Bytes



Hexadecimal

■ Base 16

- ❖ Compact representation of bit strings
- ❖ 4 bits per hex digit

0	0000	4	0100	8	1000	C	1100
1	0001	5	0101	9	1001	D	1101
2	0010	6	0110	A	1010	E	1110
3	0011	7	0111	B	1011	F	1111

A 1010₂ 10
B 1011₂ 11
C 1100₂ 12
D 1101₂ 13
E 1110₂ 14
F 1111₂ 15

- Example: ECA86420₁₆
 - 1110 1100 1010 1000 0110 0100 0010 0000₂
14 12 10 8 6 4 2 0
 - Starts with “0x” prefix: 0xECA96420

Addition & Overflow

$$\begin{array}{rcccccl} & & & \textcolor{red}{1} & & \\ & 1 & 0 & 0 & 1 & _2 \\ + & 0 & 1 & 0 & 1 & _2 \\ \hline 1 & 1 & 1 & 0 & & _2 \end{array}$$

$$\begin{array}{r}
 \begin{array}{cccccccc}
 & 1 & & 1 & & 1 & & 1 \\
 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\
 + & 1 & 1 & 1 & 1 & 1 & 0 & 1
 \end{array}
 \end{array}$$

- Digital systems operate on a fixed number of bits
- Addition overflows when the result is too big to fit in the available number of bits

Singed Binary Numbers

- How represent negative integer numbers?
 - ❖ Sign / Magnitude
 - ❖ Two's Complement

Sign / Magnitude Numbers

- 1 sign bit, N-1 magnitude bits (absolute number)
- Sign bit is the MSB (left-most bit)
 - ❖ Positive : sign bit is 0
 - ❖ Negative : sign bit is 1

- Example: 4-bit representations of +5 and -5:

$$+5 = 0101_2$$

$$-5 = 1101_2$$

- Range of an N-bit sign/magnitude number:

$$\underline{[-(2^{N-1}-1), 2^{N-1}-1]}$$

Sign / Magnitude Number Problems

- Addition doesn't work naturally
- Example: $5 + (-5) = 0$?

$$\begin{array}{r|rrrr} & 0 & 1 & 0 & 1 & _2 \\ + & 1 & 1 & 0 & 1 & _2 \\ \hline 1 & 0 & 0 & 1 & 0 & _2 \end{array}$$

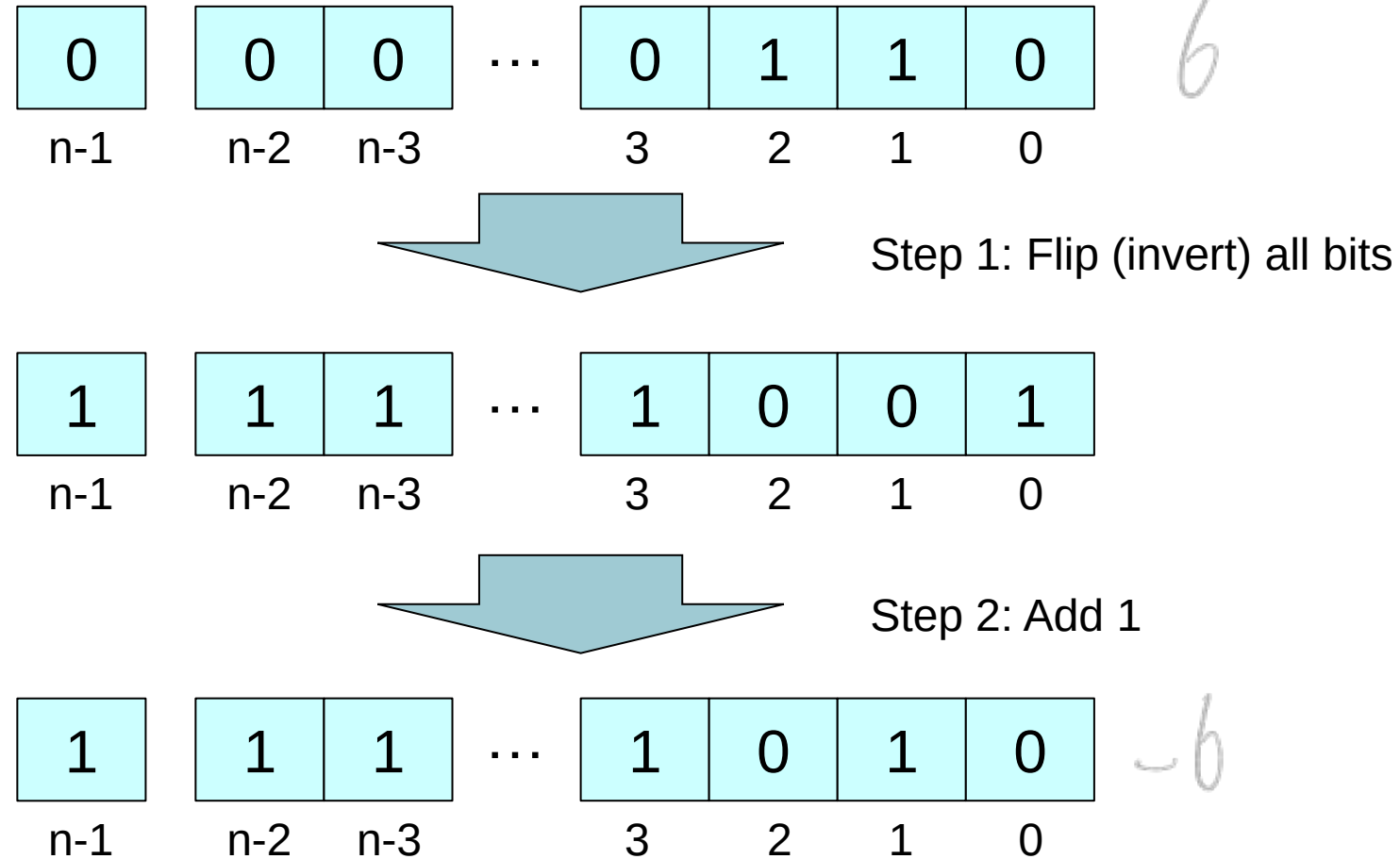
- Two representations of 0 (+0 and -0) *0이 2개의 종류*

0000 (+0)

1000 (-0)

Two's Complement

- To negate a positive integer value,



Two's Complement Addition

- Example: $5 + (-5) = 0$?

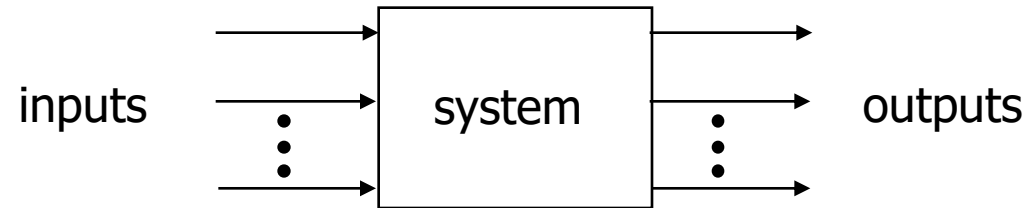
+5: 0 1 0 1₂
Step 1: 1 0 1 0₂
Step 2: 1 0 1 1₂

The diagram shows the addition of 5 and -5 in two's complement. The number 5 is represented as 0101₂ and -5 as 1011₂. They are added together, resulting in 10000₂. The leftmost '1' is the carry-out, and the remaining '0000' is the result. Arrows point from the labels '+5' and '-5' to their respective binary representations.

	0	1	0	1	₂
+	1	0	1	1	₂
1	0	0	0	0	₂

Combinational vs. sequential circuits

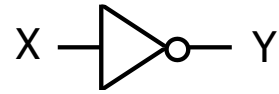
- A simple model of a digital system is a unit with inputs and outputs:



- Combinational means “memory-less”
 - ❖ Output values are determined by its current input values
- Sequential means “with memory”
 - ❖ Output values depend on the past history of input values

Combinational Logic - Logic Gates

- NOT: X'



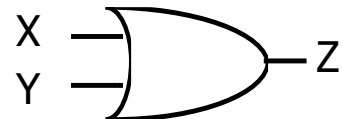
X	Y
0	1
1	0

- AND: $X \cdot Y$



X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

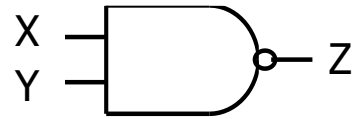
- OR: $X + Y$



X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

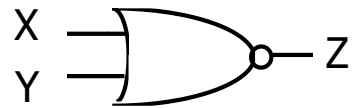
Combinational Logic - Logic Gates

■ NAND



X	Y	Z
0	0	1
0	1	1
1	0	1
1	1	0

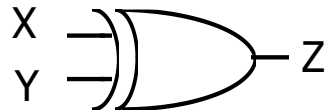
■ NOR



X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0

■ XOR

$$X \oplus Y$$



X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

■ XNOR

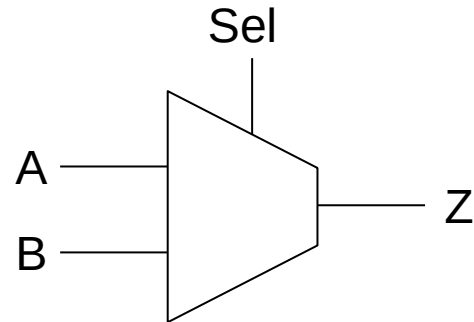
$$X = Y$$



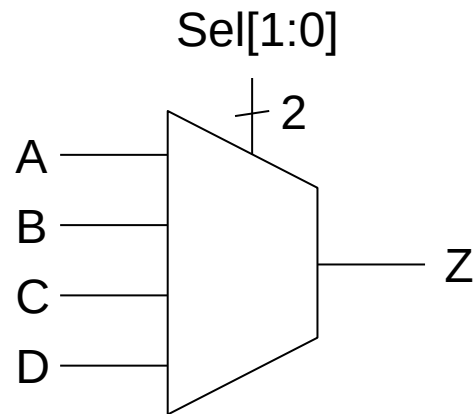
X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	1

Combinational Logic – Mux / Demux

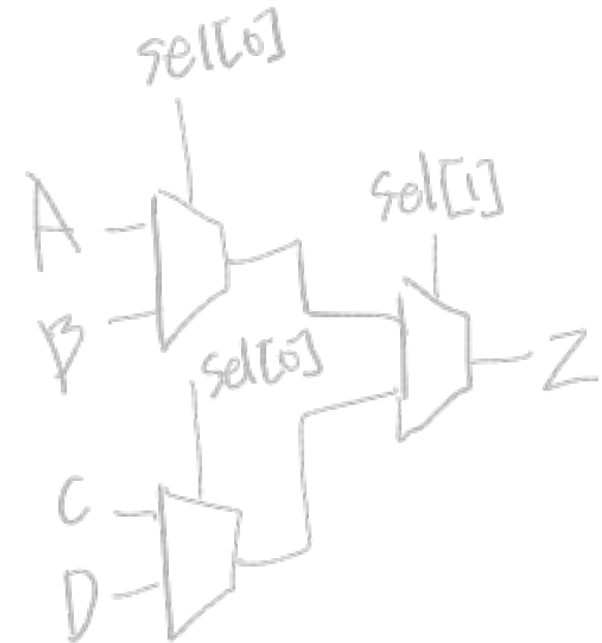
■ Mux



Sel	Z
0	A
1	B

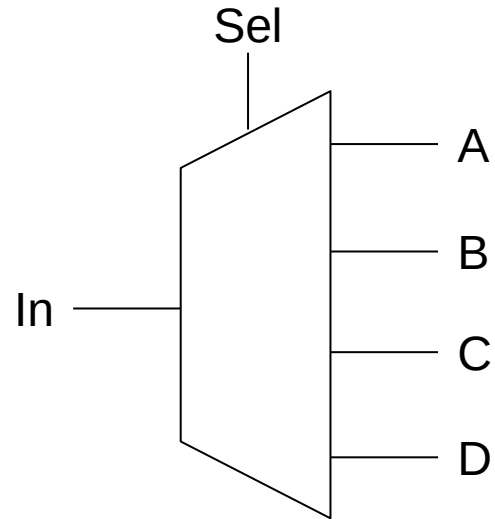


Sel	Z
00	A
01	B
10	C
11	D



Combinational Logic – Mux / Demux

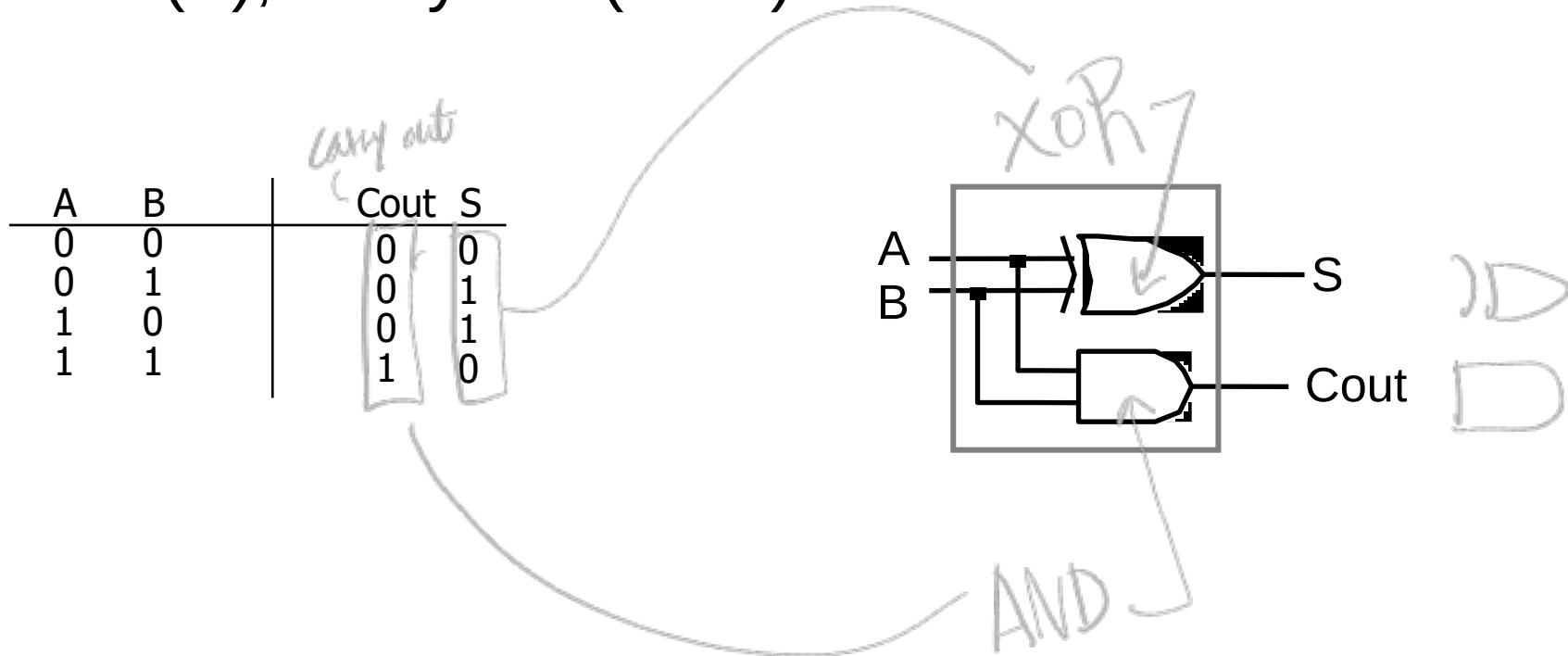
- Demux



Sel	A	B	C	D
00	In	0	0	0
01	0	In	0	0
10	0	0	In	0
11	0	0	0	In

Combinational Logic: 1-bit Binary Adder

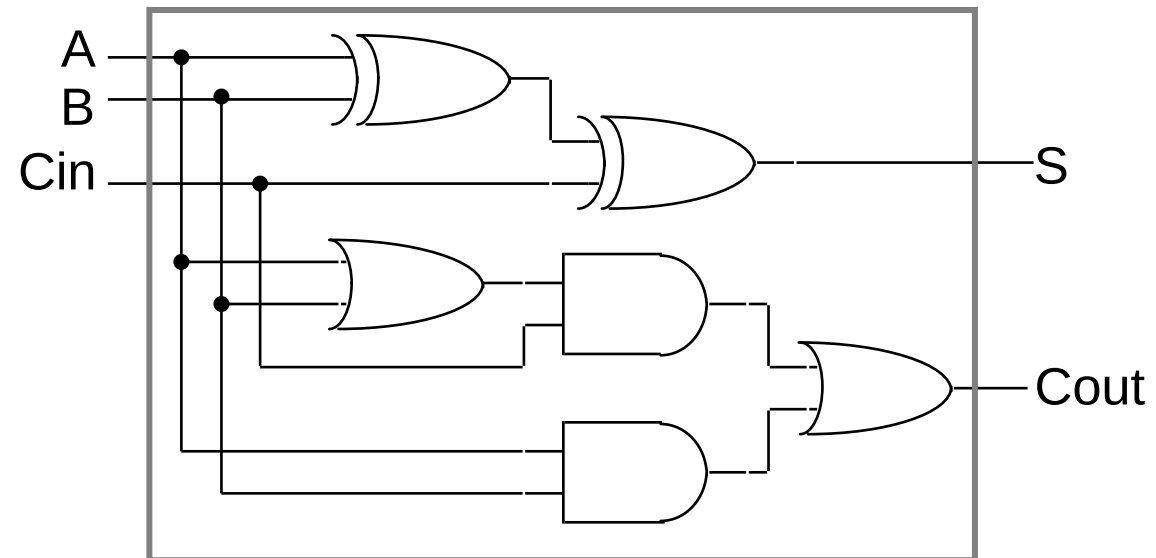
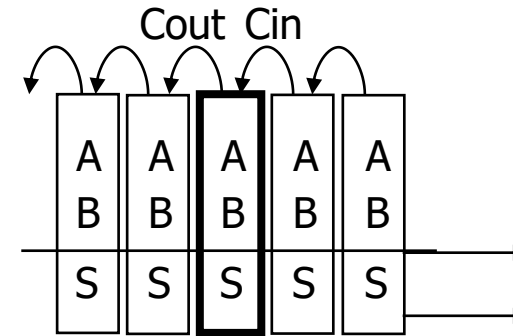
- Inputs: A, B
- Outputs: Sum (S), Carry-out (Cout)



Combinational Logic: 1-bit Binary Full Adder

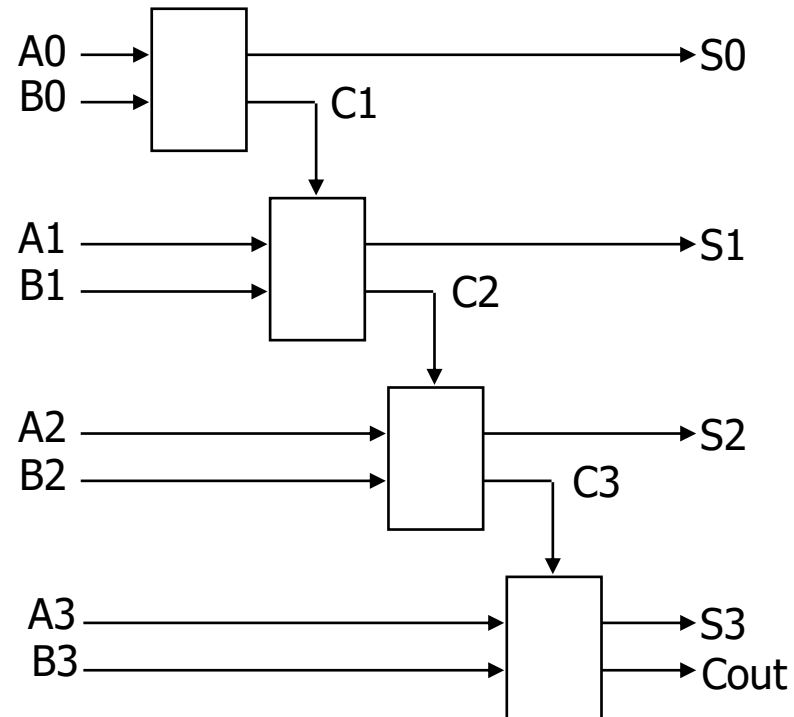
- Inputs: A, B, Carry-in
- Outputs: Sum, Carry-out

A	B	Cin	Cout	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

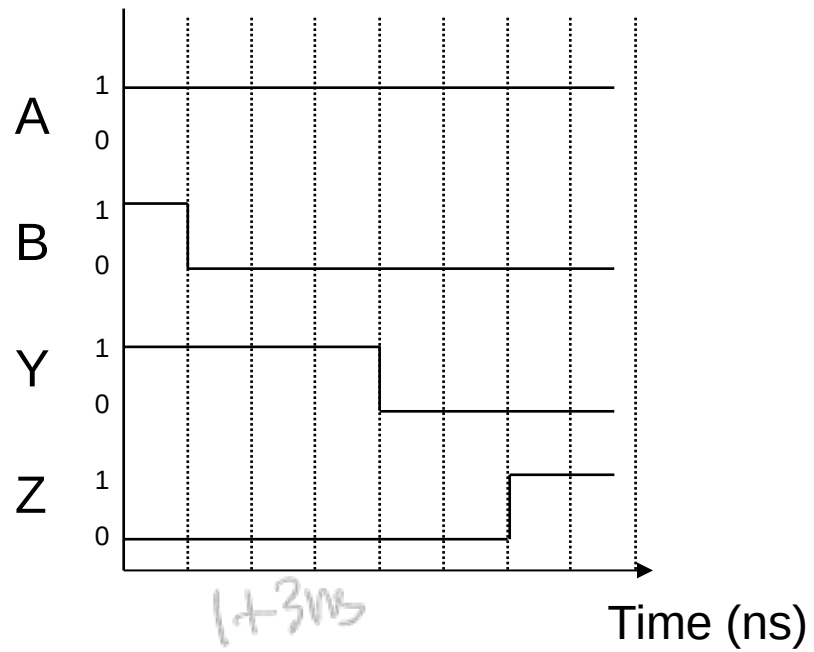
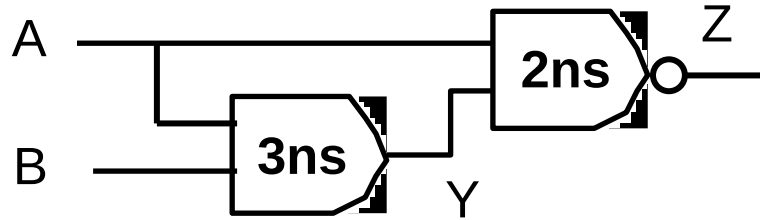


Ripple Carry Adder

$$\begin{array}{|c|c|c|c|} \hline A3 & A2 & A1 & A0 \\ \hline \end{array} + \begin{array}{|c|c|c|c|} \hline B3 & B2 & B1 & B0 \\ \hline \end{array} = \begin{array}{|c|} \hline \text{Cout} \\ \hline \end{array} \begin{array}{|c|c|c|c|} \hline S3 & S2 & S1 & S0 \\ \hline \end{array}$$



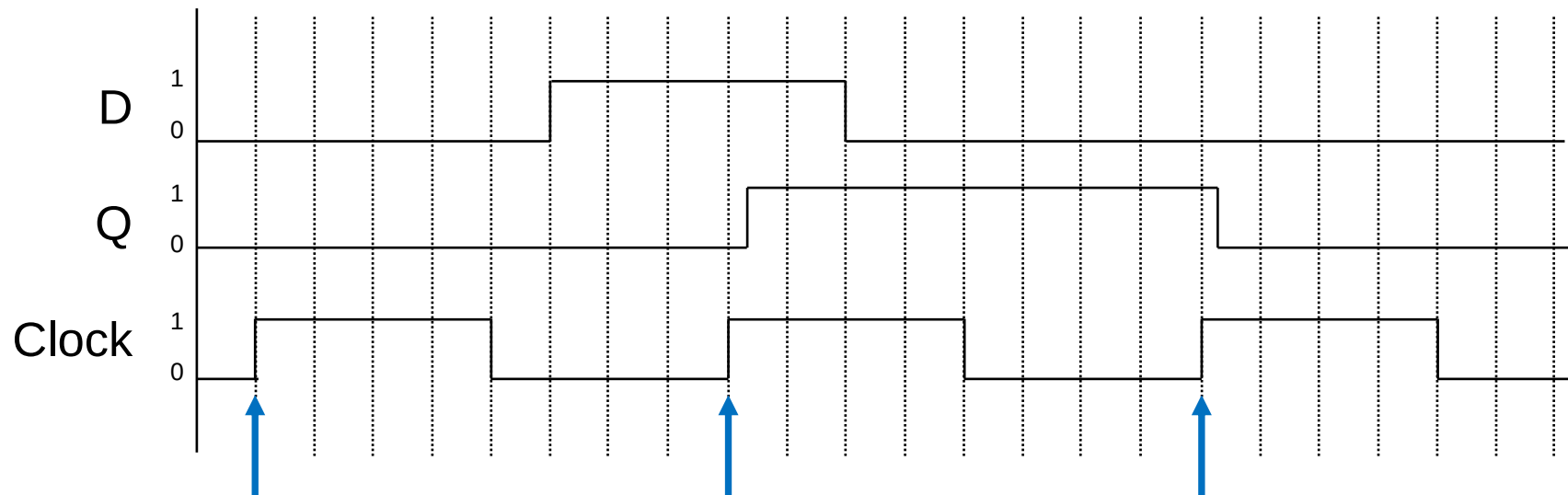
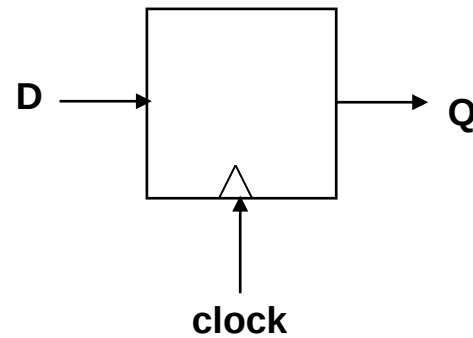
Pulse Propagation



Propagation Delay

Sequential logic

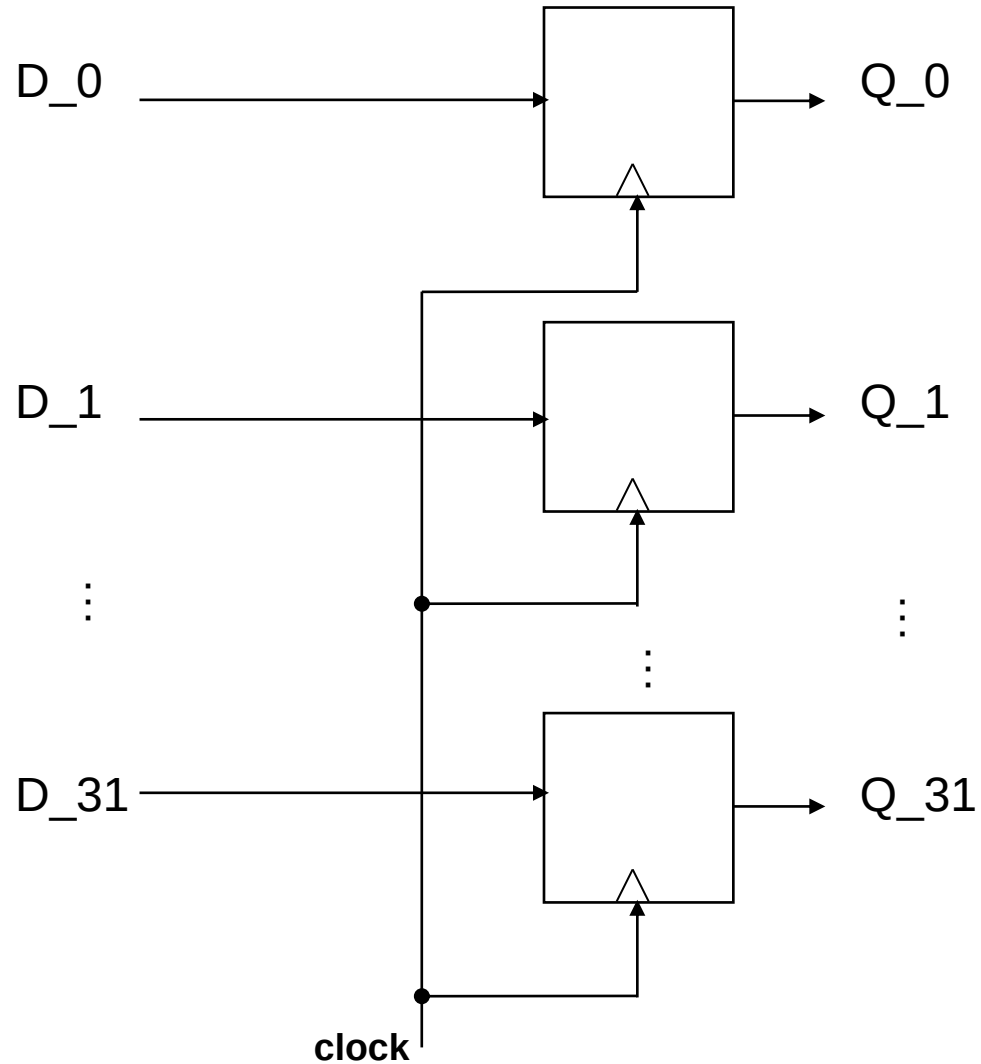
- **Storage elements** to remember the current state
- Common sequential logic elements are called **flip-flops**
 - ❖ Sense input value on rising clock edge



1GHz
 10^9

Sequential logic

- Multiple flip-flops can be used together to represent a multi-bit data



Sequential logic

- In all real circuits, outputs depend on inputs and the entire history of execution!

